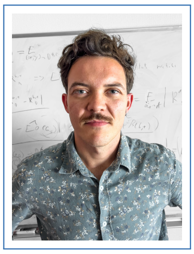
Volodimir Mitarchuk

Job title: postdoc in NLP

42100 Saint Etienne, France
+33 6 69 32 37 94
volodimir.mitarchuk@univ-stetienne.fr
github.com/23Vladymir57



Profile

Doctor in Fundamental Machine Learning with a strong background in mathematics and theoretical computer science. My current research focuses on the integration of physical knowledge within machine learning frameworks, and deriving generalization guaranties for generative models. I am currently part of the Inria MALICE research team.

Experience

2025–Present **PostDoc Position**, *MALICE Inria Team*
My research is on two fronts. 1) Application of RKBS (Reproducing Kernel Banach Spaces) theory to PIML (Physics Informed Machine Learning). 2) Generalization of generative models (in particular PAC-Bayes for Normalizing Flow models).

2025 **Reading Group Animation**, *MALICE Inria*
I led a reading group on the theme of Operator Learning for Physics Informed Machine Learning.

2024 **Internship Supervision**, *First Year international Master student*
Topic: "On the difficulty of training Saturated RNN". Exploring empirically the paths opened up by my theoretical work with a trainee targeting the learning of saturated RNNs.

2024 **Consortium Animation**, *ANR project TAUDoS*
Launch and animation of a newsletter, co-animation of workshops.

2023 **Research Visit**, *MILA Center*
A 1-month visit at the MILA center in Montreal, at the team of G. Rabusseau. Led to a publication in PAC-Bayes theory for RNN

2021 **Internship**, *Theory of Machine Learning*
Six month research internship on the topic of explainable AI. Led to the publication of "The Active Nonsmooth Manifolds of a Neural Network Classifier"

Publications

Journal **TMLR 2024**: On the theoretical limit of gradient descent for Simple Recurrent Neural Networks with finite precision.

Conference **AIStats 2024**: A PAC-Bayes bound for Recurrent Neural Nets (RNN) that does not depend on the data length.

Conference **Science and Information Conference**: The Active Nonsmooth Manifolds of a Neural Network Classifier: A Tool for Confidence Assessment.

Workshop **LearnAut 2024**: Theoretical study of incremental counting in LSTM in finite precision.

Workshop **LearnAut 2022**: Theoretical limitations of gradient descent in finite precision.

Education

2021–2025 **Ph.D. in Computer Science**, *University Jean Monnet*

- **Thesis Title:** Saturation in Recurrent Neural Networks: Expressivity, Learnability and Generalization.
- **Supervisors:** Rémi Eyraud, Amaury Habrard, and Rémi Emonet.
- **Jury:**
  - Rémi GRIBONVAL (president)
  - Hachem KADRI (rapporteur)
  - Marc TOMMASI (rapporteur)
  - Marianne CLAUSEL (examiner)
  - Rémi EYRAUD (director)
  - Amaury HABRARD (co-director)
  - Rémi EMONET (superviser)
- **Keywords** Theoretical Machine Learning, Recurrent Neural Networks, Gradient Descent, Finite Precision, PAC-Bayes Bounds, Deep Learning.
- **Description** My doctoral research focuses on understanding the theoretical limits and capabilities of Recurrent Neural Networks (RNNs), particularly in the context of finite precision. The study aims to explore the expressivity, learnability, and generalization properties of RNNs, providing insights into their practical applications and limitations.
- **Research Themes during the PhD**
  - Theoretical analysis of gradient descent in finite precision for RNNs.
  - Development of PAC-Bayes bounds for RNNs that are independent of data length.
  - Study of the expressivity and generalization capabilities of saturated RNNs.

2020–2021 **Master in Mathematics**, *ENS de Lyon*

Specialization in the mathematical foundations of machine learning.

2019–2020 **Master in Mathematics**, *Aix-Marseille University*

First year of master's degree in fundamental mathematics.

Additional Education

2025 **Spring School on Generative AI Models**, *GeMSS Statlearn, Sophia Antipolis*

An intensive one-week training course on generative models. An intensive one-week training course on generative models for researchers. These courses detailed generative models from PCA to auto-regressive models such as ChatGPT and diffusion models such as Stable Diffusion.

2022 **Spring School on Theoretical Computer Science**, *EPIT, CIRM Marseille*

A week of intensive training on all aspects of machine learning models, with a strong emphasis on theory. We covered graph learning, bandit learning, privacy, algorithm convergence guarantees, and the contribution of deep learning to NLP.

Reviewing

ICML 2025, NeurIPS 2025, LearnAut 2023, NeurIPS 2023

Technical Skills

Programming	Python
Libraries	TensorFlow, PyTorch, Scikit-learn
Fields	Machine Learning, Deep Learning, Probability Theory, Functional analysis, Linear Algebra

Personal Information

Citizenship	French
Languages	French (Native), English (Fluent), Ukrainian (Fluent), Russian (Intermediate)

Hobbies

Music	I perform live with my guitar at local rock venues.
Sports	I practice calisthenics several times a week.
DIY	I enjoy building stuff with my hands (pizza oven, car maintenance).
Transport	I am passionate about the automotive design/history.